

Assignment

A

① what is a string in c ? Define what a string is and provide a simple example of a string declaration in c.

→ Strings are used for storing text / characters. A string in c is a sequence of characters stored in a character array and terminated by a null character (~~0~~) (`\0`).

Example: "Hello" is a string of characters

- Note that you have to use double quotes (" ").
- To output the string, you can use the `printf()` function together with the format specifiers %s to tell c that we are now working with strings

- You can declare string in two ways

- Using character array:

ex `char str[6] = "Hello"`

- Without size

Ex: `char str[] = "Hello"`

- Character-by-character

ex: `char str[6] = {'H', 'e', 'l', 'l', 'o', '\0'}`

- Using pointers

`char *str = "Hello";`

Program example:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char str[20];
```



```

printf("Enter a string:\n");
scanf("%s", str);
printf("%s", str);
return 0;
}

```

output

Enter a string

Hello

Hello

② How do you find the length of a string? write a program that uses `strlen()` to find and print the length of a string.

→ The `string.h` library in C provide standard functions to work with strings. Example: `strlen()`, `strcpy()`, `strcat()`, `strcmp()`.

- C also has many useful string functions which can be useful to perform certain operations on strings.
- To use them you must include the `<string.h>` header file in your program.

String length

- For example: to get the length of a string, you can use the `strlen()` function

ex:

```
char alphabet[] = "Kaushal"
printf("%d", strlen(alphabet));
```

Program:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char name[20];
    printf("Enter your name: ");
    scanf("%s", name);
```



```
printf("length = %d\n", strlen(name));
printf("size = %d\n", strlen(-1 * sizeof(name)));
return 0;
}
```

Output

Enter your name: Kaushal

length = 87

size = 20 // the memory size (in bytes), not the
// actual string length

③ How do you copy a string in c ? Demonstrate how to use strcpy() to copy the one string to another with an example.

→ To copy the value of one string to another, you can use the strcpy() function:

Example

```
• char str1[20] = "Hello world";
```

```
  char str2[20];
```

```
// copy str1 to str2
```

```
strcpy(str2, str1);
```

```
// print str2
```

```
printf("str2: %s", str2);
```

Note: these strcpy() all function are already coded and are build in in <string.h> headerfile so it is easy to use

Program:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
  char name[20];
```

```
  char initial[20];
```



```

printf("Enter your name:");
scanf("%s", name);
printf("Copied name to initial = %s in", strcpy(initial, name));
return 0;
}

```

Output:

Enter your name : Kaushal

Copied name to initial = Kaushal

④ What is a structure? Explain what a structure is and why we use it in C. Provide a simple example of declaring a structure.

→ A structure in C is a user-defined data type that allows grouping together different data types under one name. Think of it like a container that holds multiple variables of different types, all related to a specific concept.

- Each variable in the structure is known as a member of the structure. Use the struct keyword inside the main() method to use structure.

Why we use structure

for example:

If we are storing information about a student i.e. (name, roll number, and marks) we need.

- `char name[50];`
- `int roll-number;` & without structure
- `float marks;`

- Here managing these variables separately can be error prone and inconvenient, specially doing it for multiple students. But structure make it easy

Ex: `struct student {`

`char name[50];`

`int roll-number;`

`float marks;`

`};` // we can manage all data as a single unit.

& with structure

Declaring a structure

- You declare a structure using the struct keyword. Here's the

Syntax:

```
struct student structureName {
    data-type member1;
    data-type member2;
    --
};
```

Program:

```
#include <stdio.h>
```

```
struct student {
```

```
    int rollnumber;
```

```
    char name[50];
```

```
    float marks;
```

```
};
```

```
int main()
```

```
{
```

```
    struct student student;
```

```
    printf("Enter Roll number : ");
```

```
    scanf("%d", &student.rollnumber);
```

```
    printf("Enter Name : ");
```

```
    scanf("%s", student.name);
```

```
    printf("Enter marks : ");
```

```
    scanf("%f", &student.marks);
```

```
printf("In student Details:\n");  
printf("Roll number: %d\n", student.rollnumber);  
printf("Name: %s\n", student.name);  
printf("marks: %f", student.marks);  
return;  
}
```

Output:

Enter Roll number: 100

Enter Name: Kaushal

Enter marks: 95.5

Student Details:

Roll number: 100

Name: Kaushal

marks: 95.5

Note:

- struct is the keyword to define a structure
- Student is the name of the structure
- name, rollnumber, and marks are the members of the structure.

⑤ How do you access structure elements? write a short program that declares a structure for a book (with title and author) and shows how to access and print the structure elements.

→ It is simple, to access elements of a structure, you use the dot Operator (.). Let's write a short code according to the question.

program:

```
#include <stdio.h>
struct Book {
    char title[50];
    char author[50];
};

int main()
{
    struct Book book;
    printf("Enter the title of the book:");
    scanf("%s", book.title);

    printf("Enter the author of the book:");
    scanf("%s", book.author);

    printf("\n Book Details are :\n");
    printf("Title : %s\n", book.title);
    printf("Author : %s\n", book.author);
    return 0;
}
```

Output:

Enter the title of the book: A classical approach to AI

Enter the author of the book: Dr. Munesh Chandra Tivedi

Book Details are:

Title: A classical approach to AI

Author: Dr. Munesh Chandra Tivedi

Note:

- book.title accesses the title member of book.
- book.author accesses the author member of book.